

Software Design Document

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1. Introduction

The **Road Safety Analysis System** is designed to analyze road accident data and provide insights that help authorities and users understand accident patterns. The system collects accident-related information such as location, road conditions, time of accidents, and vehicle types, and analyzes the data to identify high-risk areas and causes.

The system aims to improve road safety awareness and support data-driven decision-making for traffic authorities. By presenting the information visually through dashboards and maps, users can easily interpret accident trends and take preventive measures.

a. Design Principles Applied: -

Abstraction

Abstraction is used to hide complex implementation details and expose only the necessary functionalities to users.

For example, the **Accident Analysis Module** abstracts the complex data processing logic and only provides summarized accident statistics and insights to the user interface.

Modularity

The system is divided into independent modules such as:

- User Management Module
- Accident Data Collection Module
- Data Analysis Module
- Visualization Module
- Reporting Module

Each module performs a specific function and can be developed or modified independently.

Cohesion

Each module has a well-defined responsibility. For example, the **Data Analysis Module** only handles accident data processing and statistical analysis, which ensures high cohesion.

Low Coupling

Modules communicate through clearly defined interfaces. For example, the **UI module interacts with the analysis module through APIs**, reducing dependency between components and making the system easier to maintain.

b. High-Level Architecture

Architecture Style: Client–Server Architecture

The Road Safety Analysis System follows a **Client–Server architecture** where:

- The **client side** provides the user interface for entering accident data and viewing analysis results.
- The **server side** processes the data, performs analysis, and manages the database.

Components

Client Layer

- Web interface
- Map visualization
- User dashboard

Application Layer

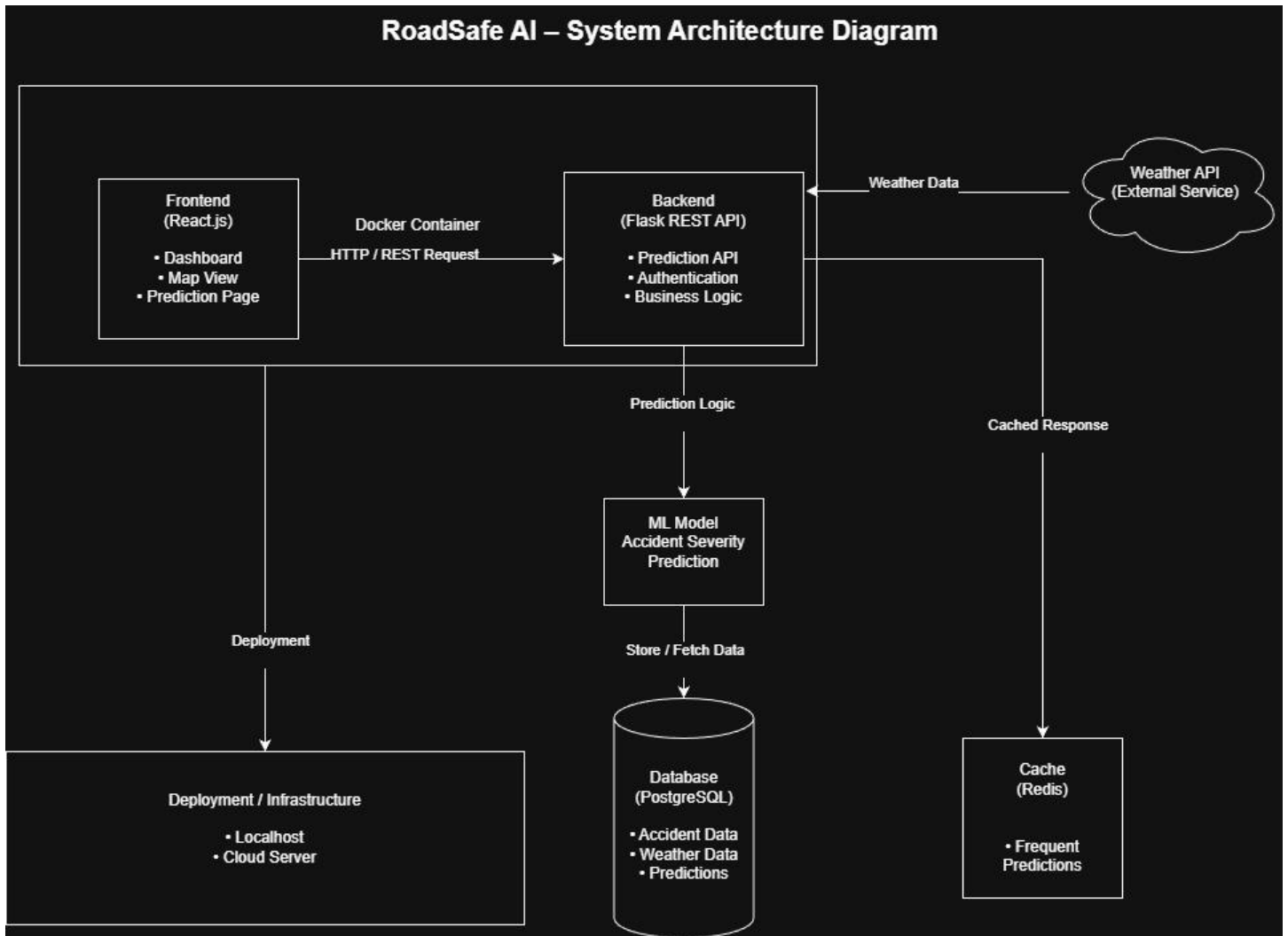
- Accident data processing
- Risk analysis algorithms
- Reporting and alerts

Data Layer

- Accident database
- User data
- Historical accident records

Why Client–Server Architecture?

- Allows centralized data storage
- Enables multiple users to access the system simultaneously
- Simplifies system maintenance and updates



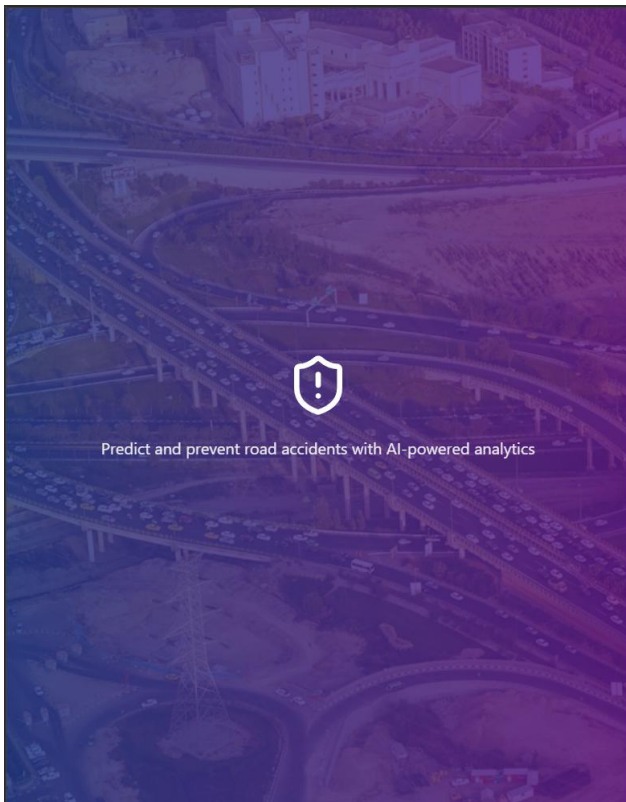
c. User Interface Design

The UI was designed using Figma to ensure a simple and intuitive user experience.

Key Screens

1. Login / Authentication Screen

- Secure access for authorized users



Sign in to your dashboard

Email Address

you@example.com

Password

☐ Remember me

[Forgot password?](#)

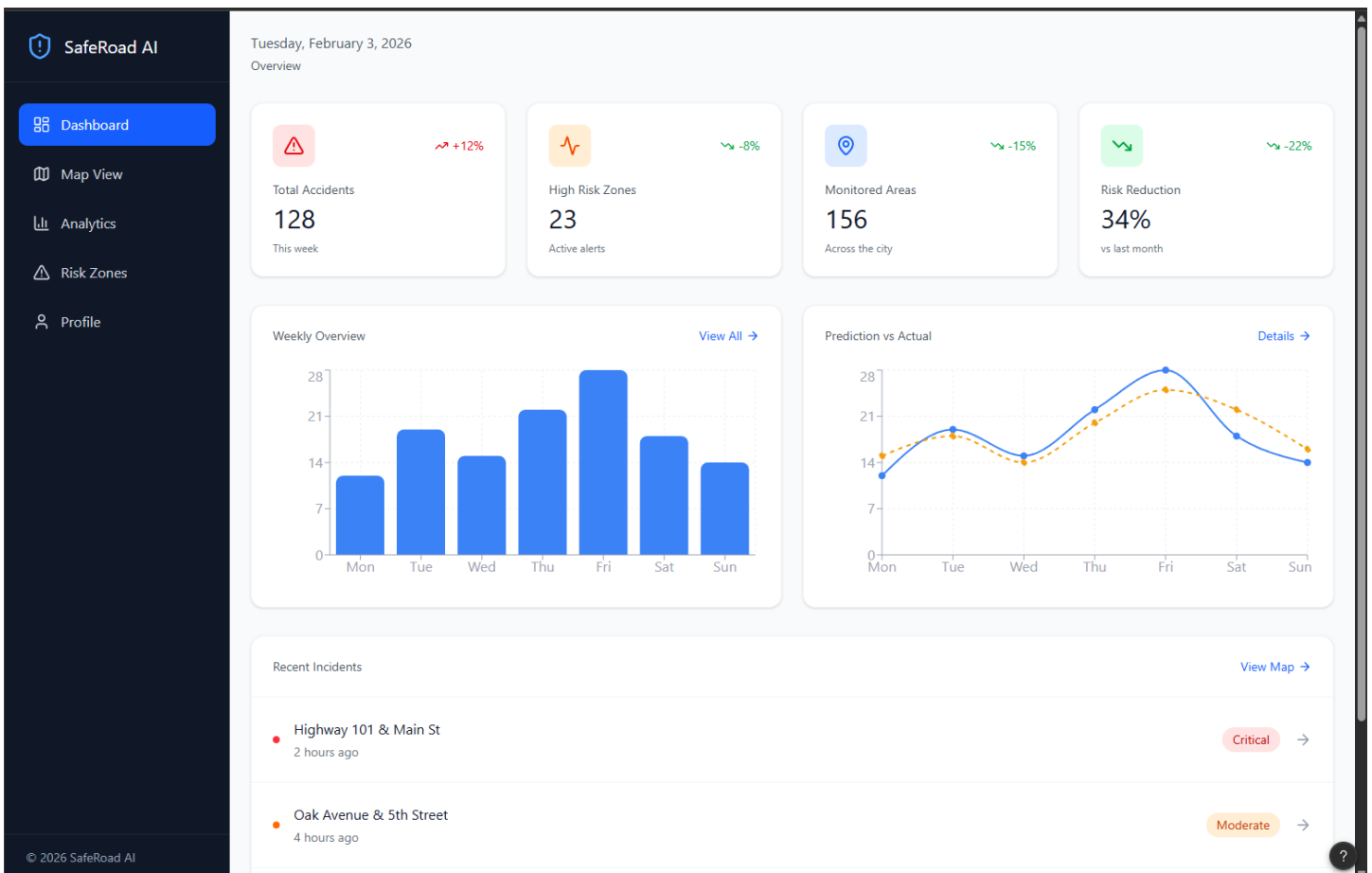
Sign In

[Don't have an account? Request access](#)



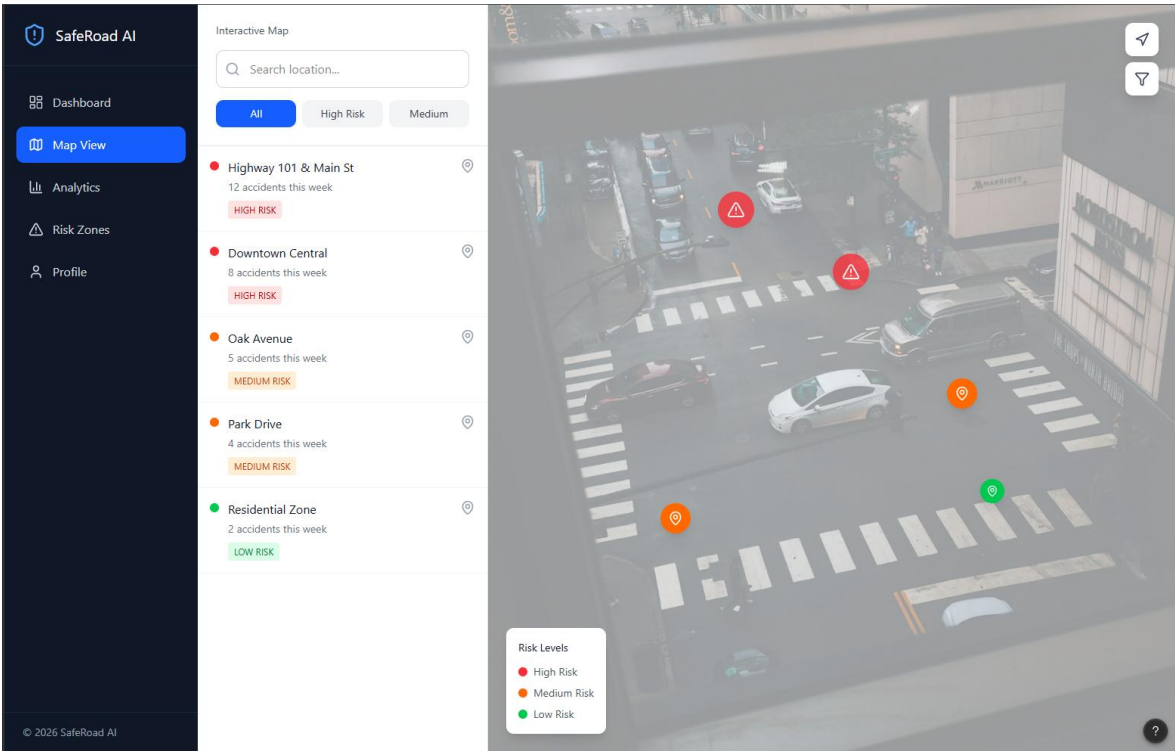
2. Dashboard

- Displays accident statistics and summary



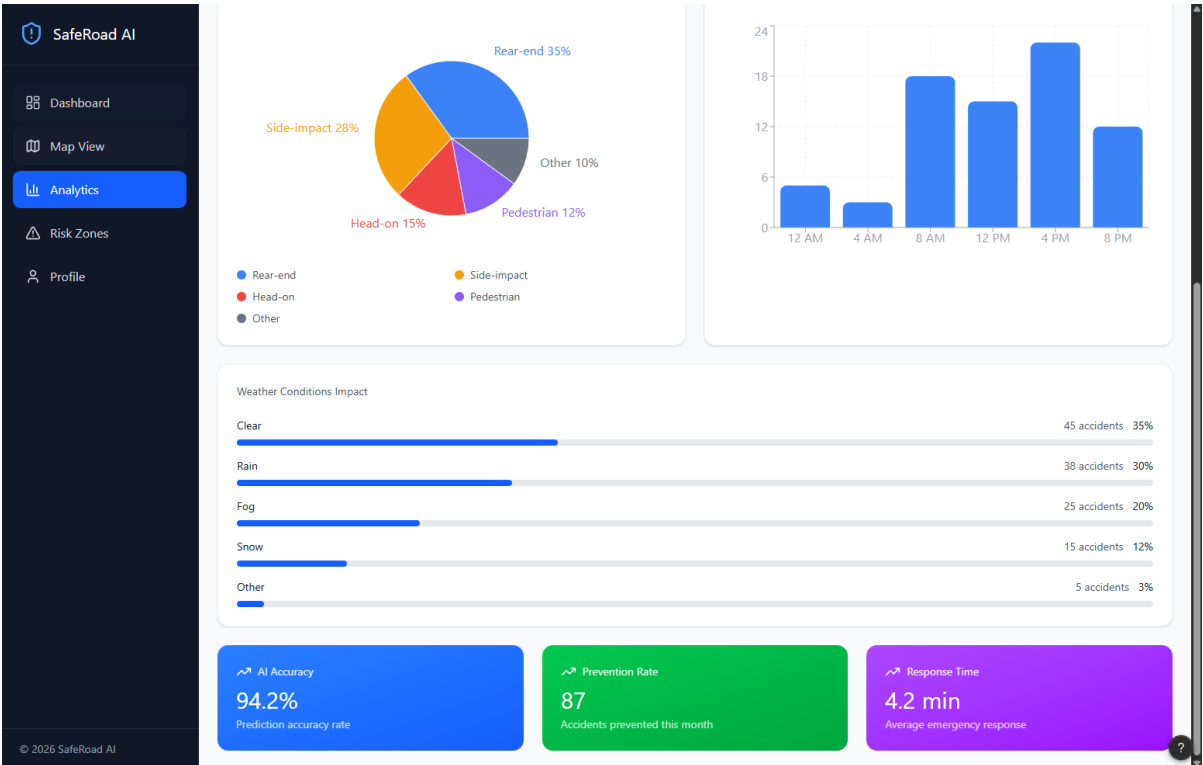
3. Accident Map View

- Displays accident locations on an interactive map



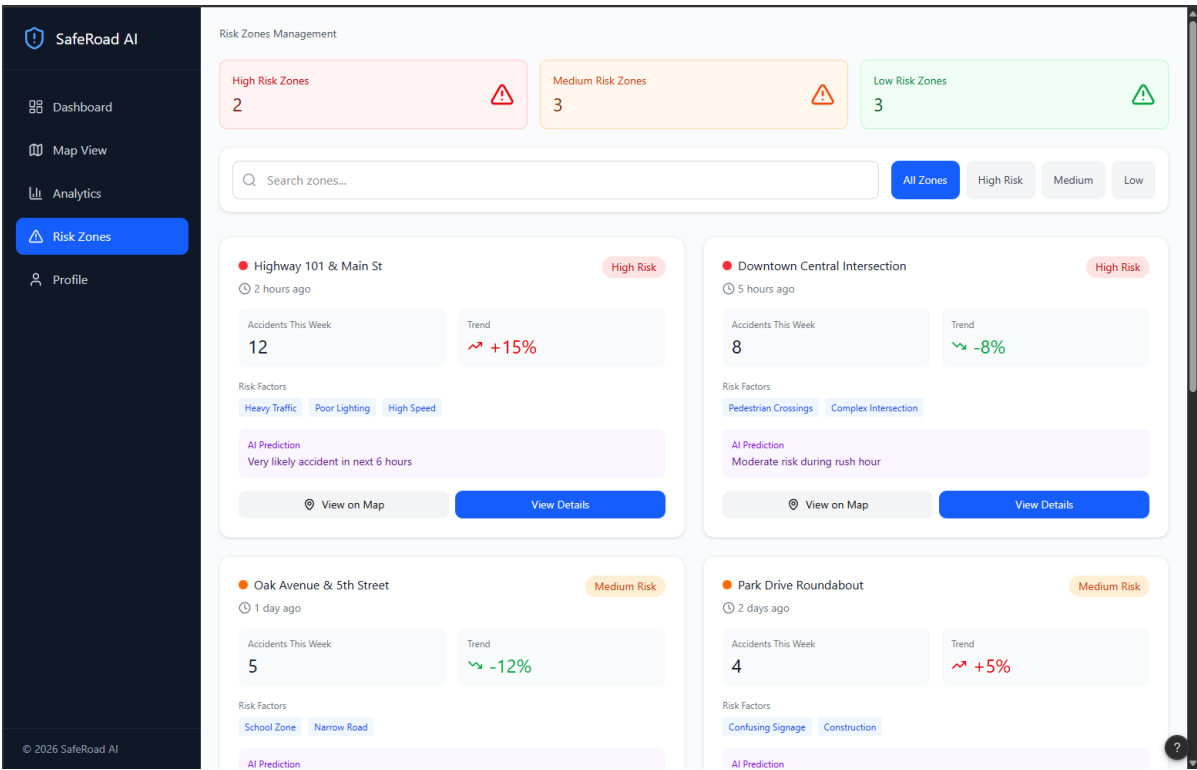
4. Analysis Dashboard

- Shows trends and accident hotspots



5. Risk zones

- Shows exact locations which are risky



6. Profile

- Users can edit their profile

The screenshot shows the 'Account Settings' page for a user named John Smith. The page is divided into two main sections: Profile Information and Notification Preferences.

Profile Information:

- Profile Information:** Includes a profile picture placeholder and the role 'Administrator' with a link to 'Update your photo and personal details'.
- First Name:** John
- Last Name:** Smith
- Email Address:** john.smith@saferoad.com
- Organization:** SafeRoad AI - City Traffic Department
- Role:** Administrator

Notification Preferences:

- Email Notifications:** Receive email updates about incidents (toggle on)
- Push Notifications:** Get push alerts for real-time updates (toggle on)
- High Risk Alerts:** Immediate alerts for critical zones (toggle on)

The left sidebar is identical to the previous screenshot, with 'Profile' now selected. The footer shows '© 2026 SafeRoad AI'.

UI Design Principles Used

- **Consistency:** Similar button styles and layout across all screens
- **Clear Navigation:** Simple menu for switching between pages
- **Visual Feedback:** Alerts and confirmations for user actions
- **Mobile Friendly Layout:** Responsive design for different devices

d. Design Decisions and Justification

Decision 1 – Separate Analysis Module

Accident analysis logic was separated from the user interface to maintain **low coupling and easier scalability**.

Decision 2 – Use Modular Architecture

Dividing the system into modules makes it easier to update individual features without affecting the entire system.

Decision 3 – Map-Based Visualization

Accident locations are displayed on maps to make the information easier to understand for users and authorities.

Decision 4 – Centralized Database

All accident records are stored in a centralized database to maintain consistency and enable efficient data analysis.

Decision 5 – Scalable Architecture

The design allows future integration with **real-time traffic data or AI prediction models**.

GitHub repository link: - [Link](#)